ML Assignment 01

Dataset: Titanic Dataset (`titanic_data_updated.csv`)
Total Marks: 100

Exam Instructions :

1. প্রথমে একটি Google Colab ফাইল খুলবে , এরপর প্রথম cell এ নিজের নাম এবং কোর্সে registration করা ইমেইল দিবে
2. Question wise numbering করে Text cell রাখবে এবং এর নিচে Code cell থাকবে, চেষ্টা করবে একটি code cell এ একটি question উত্তর দেয়ার
3. Google colab এর মধ্যে কোডগুলো করবে
4. এবং সেই ফাইলটি 'Anyone with the link' & 'View' Access দিয়ে ফাইলটির Shareble Link টি সাবমিট করবে।

Question Dataset Link:

https://drive.google.com/file/d/1aHSSJkDphOCbdGRnnXUDPHvPr08Uwtsk/view?usp=sharing

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Question 1 (10 Marks)

Load the Titanic dataset and display:

- Dataset shape
- First 10 rows
- 5 random samples

Question 2 (10 Marks)

Perform feature screening on the Titanic dataset by:

- Finding total missing values
- Finding duplicate rows
- Removing duplicate rows permanently

Question 3 (10 Marks)

Perform statistical profiling of the Titanic dataset and display:

- Dataset information
- Statistical summary of numerical columns

Also write 2 observations from the output.

Question 4 (10 Marks)

Perform univariate categorical analysis on the `Survived` column by:

- Creating a countplot
- Calculating percentage distribution
- Creating a pie chart

Write 2 insights from the analysis.

Question 5 (10 Marks)

Perform univariate numerical analysis on the `Age` column by:

- Creating a histogram
- Creating a KDE plot

Write 2 observations from the plots.

Question 6 (10 Marks)

Perform multivariate analysis between `Sex` and `Survived` using a countplot with hue.

Also calculate normalized survival ratios using `groupby()`.

Write 2 insights from the analysis.

Question 7 (10 Marks)

Create a barplot showing the relationship between `Pclass` and `Fare`.

Write 2 observations from the visualization.

Question 8 (10 Marks)

Create a KDE plot to compare the `Age` distribution of survived and non-survived passengers.

Write 2 insights from the graph.

Question 9 (10 Marks)

Perform feature engineering and train-test split by:

- Dropping `PassengerId`, `Name`, and `Ticket` columns
- Separating features (`X`) and target (`y`)
- Splitting the dataset using `test_size=0.2` and `random_state=42`

Display the shapes of train and test datasets.

Question 10 (10 Marks)

Handle missing values in the Titanic dataset by:

- Imputing missing `Age` values using `SimpleImputer` with mean strategy
- Imputing missing `Embarked` values using most frequent strategy
- Imputing missing `Cabin` values using constant strategy with `"Missing"`

Finally, check whether any missing values remain.